

Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj
B.Sc. in Computer Science and Engineering
In-Course Examination, 1st Semester - 2023
Subject: CSE-1103, Discrete Mathematics

Time: 30 Minutes

Full Marks: 20

INSTRUCTIONS

- a. Answer all the questions
- b. Figures in the margin indicate **full marks**

Student ID:

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Name:

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Question-1

04

Consider the following propositional function:

$P(x, y)$: x - y attracts each other

x and y operates over the Domain $D = \{\text{North, South}\}$ that consists of the two different poles of a magnet. It is known that different poles of magnet attract each other and the similar poles do not. Now based on the scenario, identify the truth values of the statements given below:

- $\forall_x \forall_y P(x, y)$
- $\forall_x \exists_y P(x, y)$
- $\exists_y \forall_x P(x, y)$
- $\exists_x \exists_y P(x, y)$

Question-2

06

Write down the statements of tautology for the following three rules of inference:

- Modus Tollens
- Disjunctive Syllogism
- Resolution

Consider a statistical report about Bangladesh National Cricket Team that, if Shakib Al Hasan gets a century or gets 5 wickets in a match, then Bangladesh definitely wins that match and Shakib Al Hasan becomes the Man of the match as well. Now, unfortunately, Bangladesh lost in their last match played against Ireland. Now, using appropriate rules and laws, come to the conclusion that Shakib Al Hasan could not manage a century in the last match. Note that, The name of the rules and laws must be mentioned that will be used throughout the entire proof at each step.

Question-3

06

Explain De Morgan's law for quantifiers.

Consider the following two premises.

- $\exists_x (C(x) \wedge \neg B(x))$
- $\forall_x (C(x) \rightarrow P(x))$

Now show, using appropriate rules and laws that, the premises mentioned above implies $\exists_x (P(x) \wedge \neg B(x))$ as conclusion. Note that, The name of the rules and laws must be mentioned that will be used to bring the conclusion at each step of the proof.

Question-4

04

Prove the theorem by using Contrapositive technique that “If n^2 is an odd integer then n must be odd.” After that, explain and justify why Contrapositive is a convenient technique than the Direct proof technique towards proving the theorem.